

following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time t_0 or never treated.

Specifically, we consider a setting with a random sample of N units observed over $T + 1$ time periods, where T is fixed. For each $i \in \{0, \dots, N\}$ and $t \in \{0, \dots, T\}$, we observe the outcome $Y_{i,t}$ and treatment status $D_{i,t} \in \{0, 1\}$: $D_{i,t} = 1$ if i is treated in period t and $D_{i,t} = 0$ if i is not treated in period t . Throughout we assume that the observations $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$ are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e. $D_{i,s} \leq D_{i,t}$ for $s < t$. We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with $E_i = \min \{t : D_{i,t} = 1\}$. If unit i is never treated i.e. $D_{i,t} = 0$ for all t , we set $E_i = \infty$. Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts e for $e \in \{0, \dots, T, \infty\}$, where units in cohort e are first treated at the same time $\{i : E_i = e\}$.

We define $Y_{i,t}^e$ to be the potential outcome in period t when unit i is first treated in time period e . We define $Y_{i,t}^\infty$ to be the potential outcome if unit i never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

represent the observed outcome for unit i as

$$Y_{i,t} = Y_{i,t}^{E_i} = Y_{i,t}^{\infty} + \sum_{0 \leq e \leq T} (Y_{i,t}^e - Y_{i,t}^{\infty}) \cdot \mathbf{1}\{E_i = e\}. \quad (2)$$

For any treatment that is not absorbing, if we replace the treatment status $D_{i,t}$ with an indicator for ever having received the treatment, the new treatment is absorbing by construction. Oftentimes the effect of having ever received the treatment is of interest, as it captures the path of treatment effects even though the treatment itself may be transient. For example, Deryugina (2017) is interested in the fiscal cost for a county that has been hit by a hurricane. While a hurricane itself may be transient, the impact of having had a hurricane may not be transient, hence why Deryugina (2017) codes the year of the first hurricane experienced in a county as E_i .

In the next section, we use the notation developed above to define the treatment effect of an event study design.

2.1 Defining treatment effect of an event study design

In an event study design, we define the unit-level *treatment effect* as the difference between the observed outcome relative to the never-treated counterfactual outcome: $Y_{i,t} - Y_{i,t}^{\infty}$. Recall that $Y_{i,t}^{\infty}$ denotes the potential outcome if unit i never receives the treatment. This particular counterfactual outcome $Y_{i,t}^{\infty}$ is a reasonable “baseline outcome”, though other counterfactual outcomes may be of interest as well. For example, Athey and Imbens (2018) also consider the treatment effect relative to the always-treated counterfactual outcome: $Y_{i,t} - Y_{i,t}^0$. Sianesi (2004) defines the unit-level treatment effect to be relative to the not-yet-treated counterfactual outcome: $Y_{i,t} - Y_{i,t}^e$ for $e > t$.

When dynamic treatment effects are of interest, empirical researchers commonly report the coefficient estimate $\hat{\mu}_{\ell}$ associated with indicators for being ℓ periods relative to the treatment in regression (1) as an estimate for the average lagged effect. To assess the causal interpretation of μ_{ℓ} , we need “building blocks” for its decomposition, which are the average of unit-level treatment effects at a given relative period across units first treated at time $E_i = e$, i.e. units in the same cohort e . We call this average the cohort-specific average treatment effects on the treated, formally defined below. Later in Section 3 we use them as building blocks for the interpretation of the relative period coefficients μ_{ℓ} from two-way fixed effects regressions.

Definition 1. The cohort-specific average treatment effect on the treated (CATT) ℓ periods from initial

treatment is

$$CATT_{e,\ell} = E[Y_{i,e+\ell} - Y_{i,e+\ell}^{\infty} \mid E_i = e]. \quad (3)$$

Each $CATT_{e,\ell}$ represents the average treatment effect ℓ periods from the initial treatment for the cohort of units first treated at time e . We shift from calendar time index t to relative period index ℓ which denotes the periods since treatment; for cohort e , ℓ ranges from $-e$ to $T - e$ because we observe at most e periods before the initial treatment and $T - e$ periods after the initial treatment. Relative periods allow us to compare cohorts while holding their exposure to the treatment constant.

2.2 Identifying assumptions

With the above definitions, we formalize three potential identifying assumptions for outcomes of interest in our event study design. The first assumption is a generalized form of a parallel trends assumption. The second assumption requires no anticipation of the treatment. The third assumption imposes no variation across cohorts. For each assumption, we first discuss its meaning and then compare it with similar assumptions made in the literature interpreting two-way fixed effects regressions. Later in Section 3 we interpret the relative period coefficients μ_{ℓ} from two-way fixed effects regressions under different combinations of these assumptions.

Assumption 1. (Parallel trends in baseline outcomes.) *For all $s \neq t$, the $E[Y_{i,t}^{\infty} - Y_{i,s}^{\infty} \mid E_i = e]$ is the same for all $e \in \text{supp}(E_i)$.*

If an application includes never-treated units so that $\infty \in \text{supp}(E_i)$, we need to especially consider whether these never-treated units satisfy the parallel trends assumption. Never-treated units are likely to differ from ever-treated units in many ways, and may not share the same evolution of baseline outcomes. If the never-treated units are unlikely to satisfy the parallel trends assumption, then we should exclude them from the estimation to avoid violation of this assumption.

While common in the applied literature, the parallel trends assumption is strong and oftentimes violated. For example, Ashenfelter (1978) documented that participants in job training programs experience a decline in earnings prior to the training period (Ashenfelter's dip). The timing of job training is dependent on the evolution of individual's baseline earnings, and this scenario therefore does not satisfy the parallel trends assumption. Proposition 1 is the only result in this paper derived without this assumption, but there is an

active literature studying inference under violations of the parallel trends assumption e.g. Rambachan and Roth (2020).

Our parallel trends assumption coincides with that of de Chaisemartin and D’Haultfœuille (2020). One could substitute this assumption with a different identifying assumption that baseline outcomes are mean independent of E_i i.e. at each t , $E[Y_{i,t}^\infty | E_i = e]$ is the same for all $e \in \text{supp}(E_i)$ and in particular is equal to $E[Y_{i,t}^\infty]$. This stronger assumption is plausible when the timing of treatment is indeed randomized, which is the assumption used by Athey and Imbens (2018). By taking the “fully dynamic” specification as their DGP, Borusyak and Jaravel (2017) implicitly assume this version of a parallel trends assumption. Callaway and Sant’Anna (2020a) propose a weaker version that is conditional on covariates. Finally, for a particular estimand, Goodman-Bacon (2018) identifies a weaker version that only requires a weighted average of $E[Y_{i,t}^\infty - Y_{i,s}^\infty | E_i = e]$ (averaged across cohorts) to be zero.

Assumption 2. (No anticipatory behavior prior to treatment.) *There is no treatment effect in pre-treatment periods i.e. $E[Y_{i,e+\ell}^e - Y_{i,e+\ell}^\infty | E_i = e] = 0$ for all $e \in \text{supp}(E_i)$ and all $\ell < 0$.*

Assumption 2 requires potential outcomes in any ℓ periods before treatment to be equal to the baseline outcome on average as in Malani and Reif (2015) and Botosaru and Gutierrez (2018). This is most plausible if the full treatment paths are not known to units. If they have private knowledge of the future treatment path they may change their behavior in anticipation and thus the potential outcome prior to treatment may not represent baseline outcomes. For example, Hendren (2017) shows that knowledge of future job loss leads to decreases in consumption. If the periods with anticipation behavior are known, then we may consider an alternative version of Assumption 2, which holds for pre-periods in a subset of pre-treatment periods. Depending on the application, it may still be plausible to assume no anticipation until K periods before the treatment.

The no anticipation assumption proposed by Athey and Imbens (2018) is a deterministic condition which stipulates that $Y_{i,e+\ell}^e = Y_{i,e+\ell}^\infty$ for all units i and e and $\ell < 0$. By taking the “fully dynamic” specification as their DGP, Borusyak and Jaravel (2017) allow anticipation by including pre-trends indicators in the DGP. Callaway and Sant’Anna (2020a) and Goodman-Bacon (2018) implicitly assume no anticipation by using observed outcomes in time periods before the initial treatment as the untreated potential outcomes.

Assumption 3. (Treatment effect homogeneity.) *For each relative period ℓ , $CATT_{e,\ell}$ does not depend on cohort e and is equal to ATT_ℓ .*